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SENSITIVITY-BASED OPERATIONAL MODE SHAPE NORMALIZATION: APPLICATION TO A BRIDGE

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ABSTRACT

Recently, an innovative sensitivity-based technique was introduced for the normalization of operational mode shapes purely on a basis of output-only data. The technique is based on the use of a controlled mass modification experiment and does not involve any analytical models. Moreover, it allows to extend the applicability of many modal analysis based applications towards the domain of in-operation modal testing. Previously, this method was successfully tested by means of experiments on various mechanical engineering structures. The focus of this contribution is the validation of the sensitivity-based normalization technique on a civil structure. For this purpose, measurements were performed on a bridge.

1 INTRODUCTION

During the past few years, operational modal testing proved to be a valuable alternative for the use of classic forced vibration testing. Instead of using one or more artificial excitation devices, in-operation modal testing makes use of the freely available ambient excitation caused by natural excitation sources on or near the test structure. Especially in the case of civil engineering structures, the latter can be considered as an important advantage since the use of artificial excitation devices (large shakers, drop weights) can be considered expensive and impractical ^[1]. This way, in-situ ambient vibration testing opens the way for continuous monitoring of in-operation (civil) structures ^[2, 3].

Another advantage is that the test structure remains in its operating condition during the test, which can differ significantly from laboratory conditions. A first example is given by flight flutter testing where the ground vibration testing conditions are very different from the real in-flight condition of an aircraft (temperature, aero-elastic interaction, etc.) ^[4]. Other vehicles or structures (e.g. cars, trains, off shore platforms etc.) show a similar behavior to changes in their working condition ^[5–9]. Obtaining an accurate model of a test structure in its real-life operating condition can be very important. Moreover, since all structures show a certain extent of non-linear behavior, the models obtained under real loading will be linearized for a more representative working point than during classic laboratory condition forced excitation testing.

An important drawback of operational modal analysis is that some modal parameters can no longer be determined. Since the ambient forces that excite the structure are not being measured, the modal participation factors can not be determined. Consequently, the estimated operational mode shapes are not correctly scaled since their scaling factor will depend on the unknown ambient excitation. This so-called incompleteness of the operational modal model somewhat restricts its use in certain application domains. The example of particular damage identification techniques (e.g. changes in structural flexibility method ^[10, 11], sensitivity-based techniques ^[12, 13], etc.), which require (mass) normalized mode shapes of the structure, show the need for the correct re-scaling of mode shape estimates from operational modal models. Apart from structural health monitoring, the ability to normalize the operational mode shapes allows to synthesize a 're-completed' model (e.g. Frequency Response Function (FRF) model) that can be used for many other applications: dynamical structural response prediction, force identification ^[14], vibro-acoustic applications, etc.

Until recently, no straightforward techniques were available for the normalization of operational mode shapes purely on a basis of



Figure 1: Picture of the 8-span bridge overpassing the Vomano river (a); close-up of the instrumented span (b).

output-only data. In ^[11] a number of analytical normalization techniques were compared by means of experiments performed on a bridge. All included methods involve the use of a finite element model of the structure. Since these models require a detailed knowledge of the geometry and material characteristics, their application is not always straightforward. In ^[15, 16], a new method was proposed for the regeneration of frequency response functions from output-only data on a basis of cepstrum analysis. The approach was designed for a single unknown impact force at a single location. Although the method was further extended to a general broadband excitation type ^[17], its applicability towards in-operation modal analysis is still limited due to the assumptions of a single dominant excitation force. Another method consists in performing an additional forced excitation test, in a limited number of driving points, in order to normalize the in-operational mode shape estimates ^[18]. These measurements are often made in laboratory conditions which, as mentioned before, can differ significantly from the structure's real in-operation working condition. Moreover, the use of an artificial excitation device – that can measure the applied force and adequately excite the structure – is required. In ^[19], a sensitivity-based method was introduced for the normalization of operational mode shapes on a basis of in-operation modal models only. It was shown that by adding, for instance, one (or more) masses (with well-known weights) to the test structure, the operational mode shapes can be experimentally normalized by means of the measured shift in natural frequencies between the original and mass-loaded condition. So far, the method was successfully tested on a wide range of mechanical engineering structures varying from beam and plate-like structures to an agricultural sprayer boom attached to a tractor ^[20, 21]. The main goal of this contribution is the validation of the sensitivity-based normalization technique on a full scale civil engineering structure. For this purpose, ambient and forced vibration testing experiments were performed on a bridge. The sensitivity-based normalization results were validated by comparing them to a classic driving-point normalization.

2 THEORETICAL ASPECTS

2.1 Introduction

The following sections give an overview of some of the theoretical aspects used in this paper. The computation of modal parameter sensitivity is briefly discussed in Sec. 2.2. Section 2.3 discusses the topic of sensitivity-based mode shape normalization which only uses ambient vibration (i.e. output-only) data. Some additional remarks and guidelines on the use of this method can be found in Sec. 2.4.

2.2 Sensitivity of modal parameters

For many modal analysis based applications it is often required to know how a change in one or more structural parameters (mass, stiffness or damping) will affect the modal parameters of the system. An important tool for the prediction of such changes is the

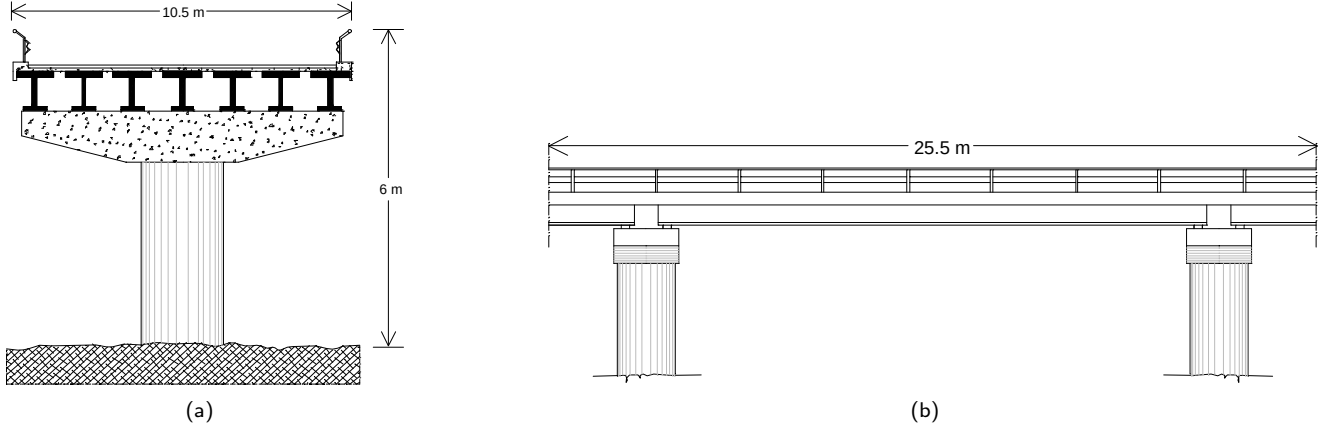


Figure 2: Schematic drawing of the instrumented span: supporting pier (a), side view (b).

knowledge of the derivatives of the modal parameters to the structural parameters in question. The poles and mode shape vectors of a linear mechanical structure are the solution (respectively eigenvalues and corresponding eigenvectors) of the eigenvalue problem associated to the equations of motion.

Many methods have been reported for the calculation of eigenvalues and eigenvectors derivatives of practically any eigenvalue problem with arbitrary matrices. An approach for real symmetric eigensystems was first presented by [22]. In [23, 24], this approach was further extended for application with more general system matrices. With these methods the determination of eigenvalue derivatives is a simple calculation since only the eigenvalue itself and the corresponding left and right eigenvectors are needed. The calculation of the eigenvector derivatives is more complicated since the knowledge of all eigenvectors of the system is required. Later work [25–27] has contributed in enhancing the efficiency since it allows the calculation of eigenvector derivatives on a basis of the associated eigenvalue and its corresponding left and right eigenvectors. However, in this case the knowledge of the system matrices is required. For a linear undamped mechanical structure, the sensitivity of a natural frequency ω_i of mode i to a local change in mass at degree of freedom (DOF) k , can be calculated by means of the estimated natural frequencies and normalized mode shapes [28–30]:

$$\frac{\partial \omega_i}{\partial m_k} = -\frac{1}{2} \omega_i \phi_{[k,i]}^2 \quad (1)$$

where $\phi_{[k,i]}$ represents DOF k of the mass normalized mode shape of mode i .

Similar expressions can be found for the sensitivity of the mode shapes and damping values to local changes in mass or stiffness. It should be noted that the actual knowledge of the system matrices is not required for the experimental computation of modal parameter derivatives. This can be considered as an advantage since building and updating accurate and reliable analytical models (e.g. a finite element model) that provide such information, is often a time consuming matter. Moreover, the experimental computation of the first order natural frequency derivative of a mode i according to (1), only requires the mode shape and natural frequency of mode i .

Sensitivity analysis proved to be extremely useful in several modal analysis application domains. A sensitivity analysis is a fast and inexpensive way of predicting the effect of structural modifications on the dynamics of a prototype without having to apply any actual high-cost changes to the structure [28–34]. As an example, a prediction of the change in natural frequency $\Delta \omega_i$, for a given mode i , induced by the presence of N small mass changes Δm_k in DOFs k of a structure can be obtained from the following first order approximation:

$$\Delta \omega_i \simeq \sum_{k=1}^N \frac{\partial \omega_i}{\partial m_k} \Delta m_k \quad (2)$$

This same expression can be used for the compensation of the mass loading effect induced by the presence of transducers and accelerometers on a test structure [35–37]. Compensating for the mass loading effect of a roving patch of accelerometers can prove necessary in order to prevent inconsistencies in data sets measured on relatively lightweight structures [38, 39].

In the following section, a technique will be introduced for the normalization of operational mode shapes on a basis of output-only data.

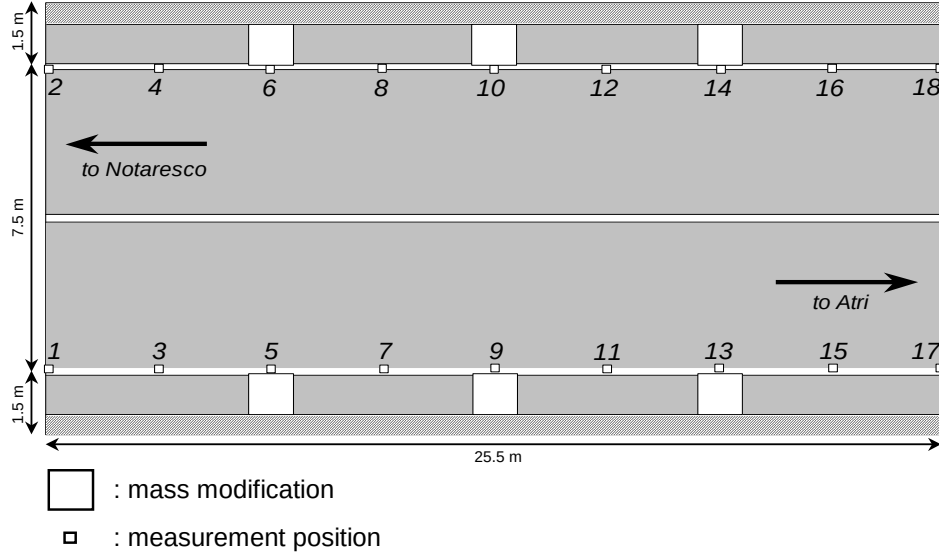


Figure 3: Measurement grid and position of the mass modifications.

TABLE 1: Schematic representation of the sensor setups: instrumented grid points 'x'.

grid point	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
setup 1	x	x			x	x			x	x			x	x			x	x
setup 2			x	x			x	x		x	x	x		x	x	x		

2.3 Sensitivity-based normalization of operational mode shapes

In ^[19], a sensitivity-based method was presented for the normalization of operational mode shape estimates obtained from output-only vibration data. The expressions for the sensitivities, such as (1), require the use of correctly normalized mode shape estimates. In case of a classical forced-vibration test, where the input forces are measured, a full modal model of the structure can be determined. As long as a driving point measurement is performed, i.e. force and structural response measured simultaneously at the same DOF, the obtained mode shape estimates can be scaled according to any normalization scheme desired ^[28]. During an in-operational modal analysis, only part of the modal model can be determined. Since the ambient forces that excite the structure are no longer being measured, the modal participation factors can not be determined. As a result, the estimated mode shape vectors remain unscaled (i.e., dependant on the unknown level and color of the ambient excitation) ^[40]. The relationship between the unscaled ($N_o \times 1$) mode shape vector $\psi_{[:,i]}$ (as obtained by in-operation modal analysis) and the corresponding correctly normalized mode shape vector $\phi_{[:,i]}$ (as can be obtained from a classical forced-vibration test) of mode i can therefore be expressed by:

$$\phi_{[:,i]} = \alpha_i \cdot \psi_{[:,i]} \quad (3)$$

with α_i an operational scaling factor for mode i dependant on the level of ambient excitation and N_o the number of outputs. It was shown in ^[19] that an estimate of the operational scaling factor can be obtained by performing a controlled mass change experiment on the test structure. By combining equations (1), (2) and (3), it is clear that a first order approximation for α_i for a mode i is given by:

$$\alpha_i \simeq \sqrt{-\frac{2\Delta\omega_i}{\omega_i \sum_{k=1}^N \psi_{[k,i]}^2 \Delta m_k}} \quad (4)$$

where k denotes the position of mass change Δm_k and N the number of mass changes used. The addition (or removal) of a small known mass Δm_k , at DOF k of the test structure, will induce a change in natural frequencies between the original and the mass loaded condition. The experimental determination of this change $\Delta\omega_i$, together with the in-operational modal model of the original structure, is sufficient to obtain an estimate of the operational scaling factor α_i .



(a)



(b)

Figure 4: View of the bridge deck with accelerometers in place (a). Picture of the truck-crane that was used to move the concrete blocks (b).

2.4 Remarks and guidelines

In order to produce accurate and reliable sensitivity-based normalization results, the following remarks and guidelines should be kept in mind:

- Placing the mass changes in or near a nodal point of the mode shape vector considered for normalization, should be avoided. The resulting change, in corresponding natural frequencies of the mass loaded and original condition, will be very small. Using such a location can produce incorrect scaling results due to the presence of uncertainties on the estimated natural frequencies.
- The accuracy of the mode shape DOFs associated with the chosen mass changes should be as high as possible. Choosing multiple mass changes ($N > 1$) can help in reducing the negative effect of outliers (i.e. mode shape DOFs with significantly higher uncertainty) among the DOFs associated with the mass changes. It should be noted that a similar remark can be made for the driving point normalization technique, where the accuracy of the driving point measurements are crucial for obtaining a reliable normalization.
- It should be noticed that the change in natural frequency $\Delta\omega_i$ can be experimentally obtained from a single measurement in a well chosen point of the structure. In some cases, e.g. multi-patch sensor setup, this can help in reducing the time between the measurements of the mass loaded and reference condition. The latter also reduces the possibility of introducing errors due to changes in natural frequency that are due to other effects than mass loading (e.g. environmental/temperature effects).
- The shift in natural frequency, between the mass loaded and unloaded reference condition, should be large enough in to allow an accurate identification. The determination of the uncertainties on the identified natural frequencies can help to assure the statistical significance of the frequency shift ^[19]. Since equation (3) is based on a first order approximation, the chosen mass

TABLE 2: Overview of the estimation results.

unloaded						loaded			comparison			
ambient			hammer			ambient						
ω (Hz)	$3\sigma_\omega$ (Hz)	ξ (%)	ω (Hz)	$3\sigma_\omega$ (Hz)	ξ (%)	ω (Hz)	$3\sigma_\omega$ (Hz)	ξ (%)	$\Delta\omega$ (Hz)	$3\sigma_{\Delta\omega}$ (Hz)	MAC	MSF
4.914	0.007	2.00	4.918	0.003	1.39	4.772	0.001	1.70	-0.142	0.008	0.9997	1.0478
5.476	0.003	2.05	5.488	0.003	1.97	5.239	0.003	1.80	-0.237	0.006	0.9992	1.0193

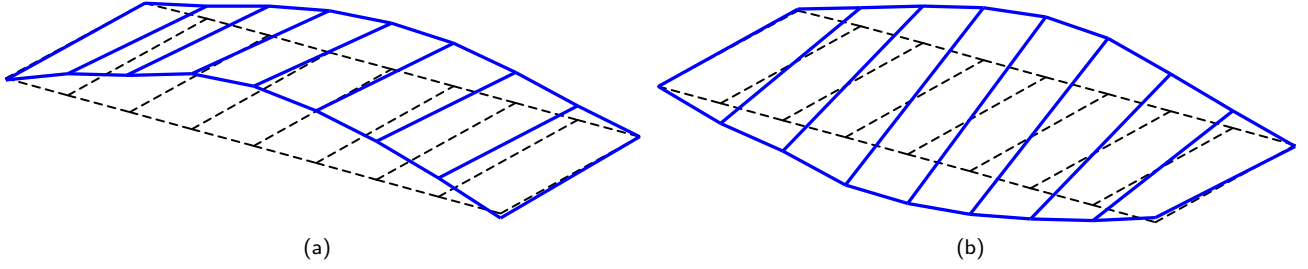


Figure 5: Mode shapes of instrumented span identified from the AVT: mode 1 (a) and 2 (b).

changes should not be too large either ^[41]. From past experience with the normalization technique, choosing the masses in a way that they produce frequency shifts of about 1 or 2 percent gives good results ^[21].

- In practice, the mass changes are generated by the addition or removal of heavy rigid objects (e.g. blocks) with finite dimensions at a certain position. It should be noted that apart from a mass change, these objects can also introduce a change in stiffness. Since the latter will produce errors on the normalization result, the dimensions of the employed mass objects should therefore be kept as small as possible.

3 EXPERIMENTAL RESULTS

In order to validate the sensitivity-based mode shape normalization procedure, vibration measurements were performed on a bridge in Italy located on the SS 553 State Highway near Fontanelle in the Teramo province. The approximately 200 m two-lane 8-span girder bridge is an overpass of the Vomano river and connects Atri to Notaresco (Fig. 1(a)). Each span is constituted of 7 main beams (prestressed reinforced concrete) and 3 orthogonal secondary beams. The supports of the bridge consist of 2 abutments and 7 piers. During the experiments, measurements were restricted to a single span supported by 2 similar piers (Fig. 1(b) and 2(a)). Previous experiments have shown that all spans have similar but independent behavior.

First of all, an ambient vibration test (AVT) was conducted under regular traffic conditions. Figure 3, gives a schematic representation of the 18-point measurement grid on the instrumented span. All vibration responses were measured in the vertical direction by means of accelerometers (Fig. 4(a)). The measured signals were sampled at 400 Hz and resulted in data sequences composed of 483328 points for each channel. Next, the periodogram method ^[42] was used for transforming the raw time-domain signals into auto and cross power spectra ready for use with a frequency-domain output-only Maximum Likelihood (ML) system identification technique ^[43]. The power spectra were computed on a basis of 116 averages allowing a 50 % overlap between the different data blocks. A Hanning window was used to reduce leakage errors.

In order to cover the entire measurement grid (experimental DOFs) with the 10 available accelerometers, two different sensor setups were used (Table 1). During the ambient vibration tests, the measurements performed at DOFs 10 and 14 were used as fixed reference response signals. The so-called reference response measurements associated with DOFs common to both patches are required during a multi-patch AVT for the purpose of eliminating the disturbing effect of the non-stationary natural excitation ^[44, 45]. Inspection of the sensor positions in Table 1 shows that the second sensor setup was mainly used for increasing the spatial resolution obtained on the span.

On a basis of the AVT data, 2 modes were clearly identified in the range from 0–10 Hz. Above this frequency range, the modes were badly excited by the traffic which made them unsuitable for further analysis. The ML estimates obtained for the natural



Figure 6: Installation of a block on the bridge deck.

frequencies (together with their 99.8% confidence intervals $3\sigma_\omega$) and damping ratios of the first two modes are given in Table 2 (unloaded/ambient). Figure 5 shows the corresponding mode shapes.

In order to correctly normalize the operational mode shapes obtained from the AVT, the sensitivity-based normalization procedure was applied to the bridge. For this purpose, a 2.3 ton mass was added to each of the 6 locations schematically represented in Fig. 3. This was accomplished by placing a 1 m^3 solid concrete cubes on the bridge deck in each of the considered locations (Fig. 7). The transportation and placement of the concrete blocks was done by means of a truck equipped with a small crane as shown in Figures 4(b) and 6. It is important to note that during all operations involved with the AVT and the sensitivity-based normalization – e.g. measuring, moving accelerometers, placing the mass modifications on the deck, etc. – the bridge remained open for traffic. In order to evaluate the mass loading effect caused by the concrete blocks, a second AVT was performed on the bridge. The measurements were performed with the same settings as during the first test. However, this time only the first sensor setup (composed of 10 measurement locations) was employed. The ML estimates of the natural frequency (together with their 99.8% confidence interval) and damping ratio of the first two modes can be found in Table 2 (loaded/ambient). The determination of the shift in natural frequency $\Delta\omega$ (converted to rad/s) allows an estimation of the operational scaling factors α_i from equation (3) for the two considered modes. Inspection of the results in Table 2 shows that both frequency shifts can be considered statistically significant with respect to the stochastic errors in the measurements (i.e. the value of $\Delta\omega$ is much larger than its 99.8% uncertainty interval $3\sigma_{\Delta\omega}$).

In order to validate the sensitivity-based normalization technique, an additional forced excitation test was performed on the bridge in its unloaded condition (i.e. no concrete blocks on the deck). The controlled artificial excitation was provided by a heavy instrumented hammer (PCB impact sledge hammer model 086D50). The experimental settings were identical to those of the previous AVTs, apart from the fact that the force signal from the instrumented hammer was recorded simultaneously with the vibration response. The bridge was excited on the deck in DOF 14, which allowed to obtain a driving point measurement and eventually a driving point normalization of the identified mode shapes. The natural frequency and damping ratio estimation results for the first two modes – obtained with the ML estimator on a basis of the computed FRFs – can be found in Table 2 (unloaded/hammer). The identified mode shapes are in good agreement with the operational mode shapes obtained from the AVT. This fact is quantitatively demonstrated by the Modal Assurance Criterium (MAC) values which are close to unity (Table 2). It is important to note that, unlike the AVT, the bridge was closed during the forced excitation test. The latter was necessary in order to reduce the errors in the input-output measurements due to the unmeasured ambient vibration. In order to keep the duration of the road obstruction as short as possible, only the first sensor setup was employed and measured.

Finally, the sensitivity-based mode shape normalization results (from output-only data) can be validated by comparing them to the results obtained from the driving point normalization (input-output data). Figure 8 shows the identified mode shape values associated with the DOFs of the first sensor setup, obtained from both the AVT (sensitivity-based normalization 'x') and the forced excitation test (driving point normalization 'o'). Both normalization results are in excellent agreement. The latter can be observed



Figure 7: Picture of the bridge deck with mass modifications in position.

more quantitatively by computing the Modal Scaling Factors ^[46] (Table 2).

4 CONCLUSIONS

In this contribution, the sensitivity-based operational mode shape normalization technique was applied and validated on a full scale bridge. This innovative method has the advantage that no analytical models of the structure are required and that it only uses data obtained by ambient vibration testing for the purpose of normalization (i.e. no measurable artificial excitation devices required). The bridge was subjected to both ambient vibration testing and a classic forced excitation test. In order to validate the normalization technique, the sensitivity-based normalization results (from the ambient vibration testing) were compared to the results obtained from a classic driving point normalization procedure that was possible due to the forced excitation test. Both methods were found to be in excellent agreement which validates the use of sensitivity-based mode shape normalization on large scale civil structures.

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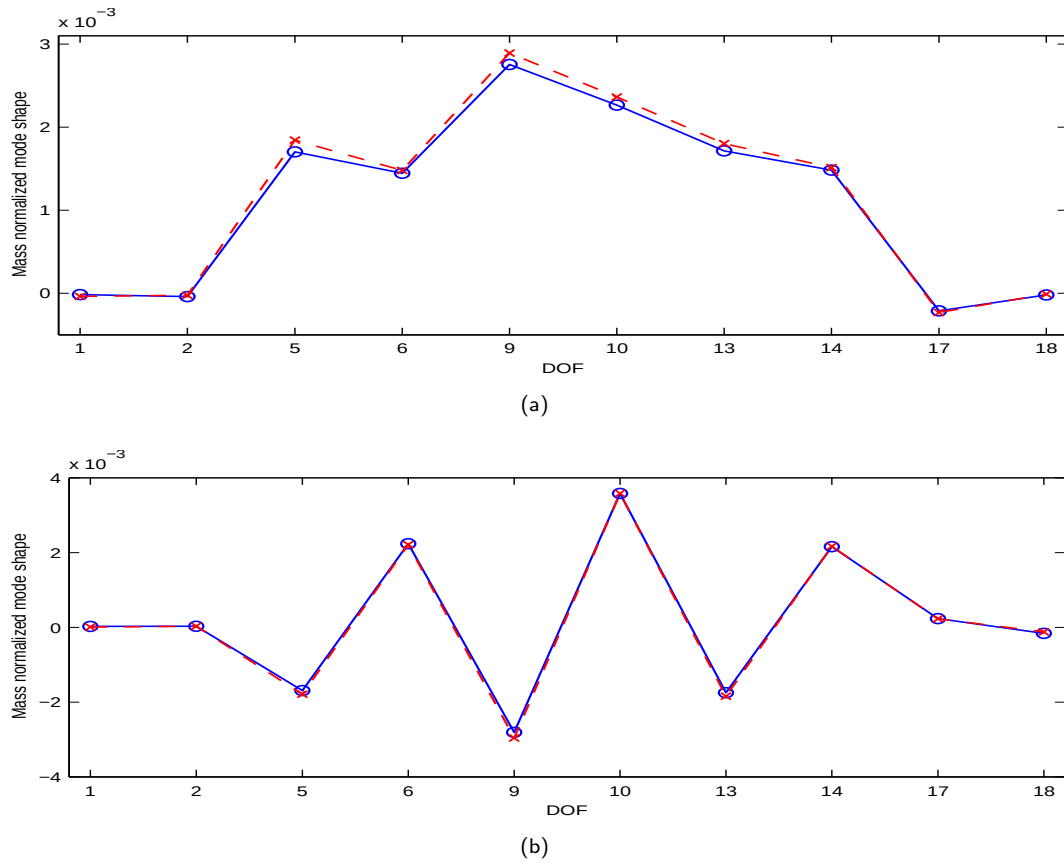


Figure 8: Comparison of the sensitivity-based 'x' and driving-point 'o' mass normalization results

lending part of the testing equipment.

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